

APPENDIX A
DFAS OF **EXAMPLE 1**

The DFA $\mathcal{E}$ described in **Example 1** which accepts $\Sigma^* \text{AddToCart}_?(x) \Sigma^* \text{AddToCart}_?(x) \Sigma^* \text{RemoveFromCart}_?(x)$, is presented in **Figure 2**. The test space defined here has only one session (0).

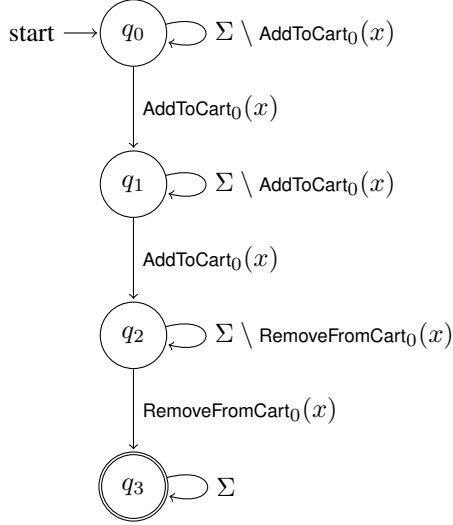


Fig. 2. An explanation DFA $\mathcal{E}$ for the bug described in **Example 1**

A minimal DFA for B for the bug described in **Example 1** is presented in **Figure 3**. It contains additional constraints from S , such as proper session initialization, login requirements, and checkout validation rules. These constraints, while important for valid test sequences, obscure the core reason for failure - the double-add followed by a remove pattern.

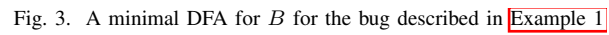


Fig. 3. A minimal DFA for B for the bug described in [Example 1](#)